

WONTHAGGI, Australia

WMO: 958810

Lat: 38.608S

Lon: 145.596E

Elev: 139

StdP: 14.62

Time Zone: 10.00 (AUS)

Period: 90-04

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	39.2	41.1	35.5	30.6	46.1	37.0	32.4	46.5	41.1	54.6	33.4	52.2	0.9	270	0.494

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	16.7	89.6	69.0	84.3	68.1	79.4	66.2	71.3	83.5	69.3	80.4	67.4	77.8	10.7	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.8	102.5	76.5	65.5	94.7	73.8	63.7	88.7	72.0	35.1	83.6	33.4	80.3	32.0	77.7	94.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 33.1	27.1	22.3	DB	34.3	99.4	2.1	6.3	32.8	103.9	31.6	107.5	30.4	111.0	28.9	115.6	(4)
(5)			WB	33.9	76.3	1.8	2.7	32.6	78.2	31.5	79.8	30.5	81.3	29.2	83.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	57.3	65.2	66.7	62.7	58.3	54.5	51.5	50.1	50.9	52.6	55.8	58.2	61.4	(6)
(7)		DBStd	7.16	5.85	6.74	5.05	4.62	3.47	3.23	2.97	3.33	4.08	4.66	5.03	5.75	(7)
(8)		HDD50	98	0	0	0	1	3	19	35	26	12	1	1	0	(8)
(9)		HDD65	3088	67	50	108	211	326	404	462	437	373	289	215	146	(9)
(10)		CDD50	2752	471	466	393	249	142	66	37	54	90	182	247	355	(10)
(11)		CDD65	268	73	96	37	9	0	0	0	0	1	5	11	36	(11)
(12)		CDH74	2404	626	847	311	55	0	0	0	0	4	65	124	372	(12)
(13)		CDH80	949	264	364	107	8	0	0	0	0	0	11	42	153	(13)
(14)	Wind	WSAvg	7.3	8.3	7.1	6.7	5.9	5.8	6.8	6.9	8.0	8.6	7.6	8.0	8.4	(14)
(15)	Precipitation	PrecAvg	37.6	2.1	1.7	2.3	3.0	4.0	3.8	4.3	4.1	3.7	3.5	2.8	2.4	(15)
(16)		PrecMax	54.3	5.9	7.4	7.5	7.5	10.8	6.9	8.2	8.5	8.6	6.5	5.4	5.6	(16)
(17)		PrecMin	24.4	0.2	0.0	0.1	0.7	1.3	1.5	1.3	1.1	0.7	1.1	0.6	0.3	(17)
(18)		PrecStd	6.7	1.1	1.6	1.3	1.7	1.9	1.5	1.9	1.6	1.6	1.5	1.3	1.3	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	97.2	96.8	90.0	79.4	71.0	64.5	61.5	67.2	73.5	80.6	86.2	92.5	(19)
(20)			MCWB	69.3	69.9	68.2	62.1	59.4	56.8	53.3	57.1	61.1	64.2	67.7	70.5	(20)
(21)		2%	DB	87.4	89.5	81.8	74.0	66.6	60.9	58.7	62.3	67.8	74.2	76.9	85.2	(21)
(22)			MCWB	69.0	69.5	66.7	61.7	57.9	55.3	52.2	54.1	57.9	61.6	64.9	66.6	(22)
(23)		5%	DB	80.9	83.7	76.8	69.9	63.5	58.8	56.9	59.7	63.9	69.0	71.5	77.7	(23)
(24)			MCWB	68.1	68.6	64.9	60.6	57.0	53.9	51.4	52.9	55.7	59.2	62.9	65.0	(24)
(25)		10%	DB	75.5	78.1	71.8	66.0	61.1	57.0	55.3	57.3	60.4	64.4	67.1	72.0	(25)
(26)			MCWB	66.0	67.0	63.7	59.2	55.9	52.7	50.5	51.5	54.3	57.5	60.4	63.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	73.5	76.0	70.7	66.8	61.8	58.3	55.3	59.2	62.5	66.3	69.9	71.8	(27)
(28)			MADB	85.2	90.5	83.8	74.3	67.2	62.7	58.4	65.7	69.6	77.1	79.5	85.7	(28)
(29)		2%	WB	70.8	71.7	67.9	63.5	59.4	56.4	53.7	55.6	60.0	62.7	66.4	68.2	(29)
(30)			MADB	83.7	82.8	77.6	70.8	63.9	59.6	57.0	60.7	67.0	71.4	76.3	81.3	(30)
(31)		5%	WB	68.5	69.6	65.7	61.4	58.0	54.7	52.4	53.6	56.9	60.2	63.3	66.0	(31)
(32)			MADB	79.2	79.9	75.3	67.7	62.3	57.8	55.6	58.4	61.7	67.9	70.5	76.2	(32)
(33)		10%	WB	66.3	67.6	63.7	59.6	56.5	53.2	51.2	52.0	54.8	57.9	60.8	63.7	(33)
(34)			MADB	75.0	77.9	71.1	64.9	60.6	56.5	54.4	56.5	59.5	63.8	66.8	71.8	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	15.6	16.7	14.2	13.8	11.1	9.5	10.1	11.1	12.0	13.6	14.1	14.7	(35)
(36)			MCDBR	27.3	27.9	25.0	21.9	15.6	13.2	12.4	15.9	19.1	23.4	24.4	28.7	(36)
(37)			MCWBR	13.1	13.1	12.1	11.3	9.9	8.9	8.1	9.9	12.0	13.2	14.9	14.6	(37)
(38)		5% WB	MCDBR	24.0	23.1	22.0	18.6	14.0	10.8	11.6	14.1	16.8	21.5	22.6	23.1	(38)
(39)			MCWBR	13.2	13.0	11.9	10.9	9.9	8.1	8.5	9.7	11.5	13.4	14.6	13.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.358	0.358	0.339	0.332	0.316	0.309	0.307	0.320	0.345	0.343	0.349	0.345	(40)	
(41)		taud	2.490	2.501	2.553	2.550	2.577	2.580	2.581	2.539	2.462	2.484	2.492	2.522	(41)	
(42)		Ebn at Noon	309	301	294	275	261	253	261	276	287	302	310	315	(42)	
(43)		Edh at Noon	36	34	30	26	22	20	22	26	32	34	36	35	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2224	1938	1525	1058	698	557	627	891	1275	1711	2011	2233	(44)	
(45)		RadStd	136	99	84	54	36	38	38	26	44	78	113	108	123	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (32 neighbors)		+0.90	N/A	N/A	+1.42	N/A	N/A	-29	-238	+311	+103

Nomenclature: See separate page